FINANCIAL DATA SCIENCE

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Some introductory remarks

- together we will do
  - linear regression - Hastie, Tibshirani, ...
  - factor models;
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MODERN LINEAR REGRESSION THEORY

An introduction to modern regression theory: the OLS estimator

Consider the linear regression model

$$y_i = \underbrace{\beta_1 + \beta_2 x_{2i} + .. + \beta_k x_{ki}}_{\text{model}} + \underbrace{\epsilon_i}_{\text{error}} \tag{1}$$

where

- the model is linear in the β_js - a model like this

  $$y_i = \beta_1 + \beta_2 x_{2i}^3 + .. + \beta_k x_{ki} + \epsilon_i$$

  is still linear;

- in this model, we try to predict y_i

  $$y_i = \underbrace{(\beta_1 + \beta_2 x_{2i} + .. + \beta_k x_{ki})}_{\text{model}} + \epsilon_i$$

  - y_i is called the *dependent variable*, or the variable to be predicted;
  - the model, $(\beta_1 + \beta_2 x_{2i} + .. + \beta_k x_{ki})$, is our predictor
    - the predictors, x_{2i}, ... x_{ki}, are often called *regressors*;
    - the quantities $\beta_1, \beta_2, ...$ are called *parameters*, slopes, coefficients, etc...
  - the perfect prediction does not exist: there will always be a discrepancy between model and reality

    $$\epsilon_i = y_i - (\beta_1 + \beta_2 x_{2i} + .. + \beta_k x_{ki}) \begin{array}{ll} > 0 & \text{understating } y_i \\ < 0 & \text{overstating } y_i \end{array}$$

    we call ϵ_i the *error term*

- the number of parameters to estimate is called/denoted as k
- we have $1 \leq i \leq N$ observations

Model (1)

$$y_i = \beta_1 + \beta_2 x_{2i} + .. + \beta_k x_{ki} + \epsilon_i$$

is used for predictions. Can we interpret this model though? Yes, with two strategies

1. to interpret $\beta_2, \beta_3, ...$ - i.e. the coefficients attached to a regressor $2 \leq j \leq k$

   $$\frac{\partial y_i}{\partial x_{ji}} = \beta_j$$

   The interpretation is "standard": if you shock x_{ji} by one of its units of measurement, then y_i will change by β_j of its units of measurement. Note that this is a partial derivative, so all the other regressors stay constant (*ceteris paribus*). Example

   $$\underset{\text{number of biscuits}}{y_i} = \beta_1 + \beta_2 \underset{\text{flour}}{x_{2i}} + .. + \beta_k \underset{\text{milk}}{x_{ki}} + \epsilon_i$$

   Then, $\beta_2 = 2.4$ means that if you increase the amount of flour x_{2i} by one kg, then you'll bake 2.4 more biscuits - note that all the other ingredients have not been modified. Finally, sometimes we use logs of our data

   $$\begin{aligned} y_i &= \ln(Y_i) \\ x_{ji} &= \ln(X_{ji}) \end{aligned}$$

   in this case,

   $$\beta_j = \frac{\partial y_i}{\partial x_{ji}} = \frac{\partial \ln(Y_i)}{\partial \ln(X_{ji})} = \frac{\frac{\partial Y_i}{Y_i}}{\frac{\partial X_{ji}}{X_{ji}}} = \frac{\partial Y_i}{\partial X_{ji}} \frac{X_{ji}}{Y_i}$$

   which is called *elasticity* of Y_i w.r.t. X_{ji}. Thus, $\beta_j = 2.4$, then it means that if X_{ji} increases by 1%, then Y_i increases by 2.4%

2. to interpret the constant/intercept β_1, I suggest that you consider the scenario where $x_{2i} = x_{3i} = ... = x_{ki} = 0$, e.g.

   $$\underset{\text{consumption}}{C_i} = \beta_1 + \beta_2 \underset{\text{income}}{I_i} + \epsilon_i$$

   you set $I_i = 0$. This scenario has its own interpretation. Then, under this scenario, $C_i = \beta_1$ - e.g. if $\beta_1 > 0$ there is consumption even if you do not have an income.

Let us go back to our model

$$y_i = \beta_1 + \beta_2 x_{2i} + .. + \beta_k x_{ki} + \epsilon_i$$

this is written in scalar form. We may write the interim form

$$\underset{k\times 1}{x_i} = \begin{bmatrix} 1 \\ x_{2i} \\ . \\ x_{ki} \end{bmatrix}; \beta = \begin{bmatrix} \beta_1 \\ . \\ . \\ \beta_k \end{bmatrix}$$

then

$$y_i = \beta' x_i + \epsilon_i.$$

Further, we can write the model in matrix form

$$\underset{N\times 1}{y} = \begin{bmatrix} y_1 \\ . \\ . \\ y_N \end{bmatrix}; \epsilon = \begin{bmatrix} \epsilon_1 \\ . \\ . \\ \epsilon_N \end{bmatrix}$$

and

$$\underset{\substack{\text{design matrix} \\ N\times k}}{X} = \begin{bmatrix} 1 & x_{21} & . & x_{k1} \\ 1 & x_{22} & . & x_{k2} \\ . & & & \\ & & & \\ 1 & x_{2N} & & x_{kN} \end{bmatrix}$$

then we have

$$\underset{N\times 1}{y} = \underset{N\times k}{X} \underset{k\times 1}{\beta} + \epsilon. \tag{2}$$

OLS estimation of β

We need to have an estimate of β. The OLS proposes this: consider

$$\epsilon_i = y_i - \beta' x_i$$

is a prediction error, and among other things, it depends on β. For a given numerical value of β, say $\widehat{\beta}$, we will have

$$\underset{\text{residual}}{\widehat{\epsilon}_i} = y_i - \underset{\text{feasible prediction, fitted value}}{\widehat{\beta}' x_i}.$$

Then I can write, in matrix form

$$\widehat{\epsilon} = \begin{bmatrix} \widehat{\epsilon}_1 \\ \\ \\ \widehat{\epsilon}_N \end{bmatrix} = y - X\widehat{\beta}.$$

We would like this to be small; how do we measure the size of a vector like $\widehat{\epsilon}$? A possible way is the L_p-norm

$$\|\widehat{\epsilon}\|_p = \left(\sum_{i=1}^{N} |\widehat{\epsilon}_i|^p \right)^{1/p}.$$

The OLS (= ordinary least squares) worries about the L_2-norm, or Euclidean distance

$$\|\widehat{\epsilon}\|_2^2 = \left\| y - X\widehat{\beta} \right\|_2^2 = \sum_{i=1}^{N} \widehat{\epsilon}_i^2 = \sum_{i=1}^{N} \left(y_i - x_i'\widehat{\beta} \right)^2 = RSS\left(\widehat{\beta}\right),$$

where RSS = residual sum of squares. Then, it is natural to define

$$\widehat{\beta} = \arg\min RSS.$$

Shall we compute this guy? Let us see some passages

$$\begin{aligned}
& \|\widehat{\epsilon}\|_2^2 \\
= \ & \widehat{\epsilon}'\widehat{\epsilon} \\
= \ & \left(y - X\widehat{\beta} \right)' \left(y - X\widehat{\beta} \right) \\
= \ & y'y + \widehat{\beta}'X'X\widehat{\beta} - y'X\widehat{\beta} - \widehat{\beta}'X'y \\
= \ & y'y + \widehat{\beta}'X'X\widehat{\beta} - 2y'X\widehat{\beta}
\end{aligned}$$

And now let's minimise. You may want to know that

$$\begin{aligned}
\frac{\partial Ax}{\partial x} &= A' \\
\frac{\partial x'Ax}{\partial x} &= \left(A + A' \right) x
\end{aligned}$$

then

$$\frac{\partial}{\partial \widehat{\beta}} \|\widehat{\epsilon}\|_2^2 = 0$$

is the first order condition. This is

$$\begin{aligned}
& \frac{\partial}{\partial \widehat{\beta}} \left(y'y + \widehat{\beta}'X'X\widehat{\beta} - 2y'X\widehat{\beta} \right) \\
= \ & 0 + 2X'X\widehat{\beta} - 2X'y = 0.
\end{aligned}$$

Then we finally have the OLS

$$\widehat{\beta} = \left(X'X \right)^{-1} \left(X'y \right). \tag{3}$$

But is this really the OLS? We should check the second order condition

$$\frac{\partial}{\partial \widehat{\beta}'} \frac{\partial}{\partial \widehat{\beta}} \left(y'y + \widehat{\beta}'X'X\widehat{\beta} - 2y'X\widehat{\beta} \right) = 2X'X$$

and this is a positive definite matrix (it looks like a square, which is never negative).

Properties of the OLS

Recall

$$\begin{aligned} y &= X\beta + \epsilon \\ \widehat{\beta} &= \left(X'X\right)^{-1}\left(X'y\right) \end{aligned}$$

We want to assess the quality of the OLS estimator.

Before we do it, note that the OLS estimator may not exist: if $X'X$ is not invertible, OLS does not exist. This means that OLS exists IFF $X'X$ is invertible, or IFF

$$rank\left(X'X\right) = k < N.$$

The OLS cannot exist when

1. $k > N$
2. $k < N$ but $rank\left(X'X\right) < k$, called *collinearity*.

In these cases, you will need another estimator.

If OLS exists, let us study its properties. The properties are spelt out in terms of the estimation error

$$\widehat{\beta} - \beta = \left(X'X\right)^{-1}\left(X'\epsilon\right).$$

Each property - formally - is a theorem, so we need some assumptions.

Let's see:

Unbiasedness Assume that

$$E\left(\epsilon\right) = 0, \tag{4}$$

or if you prefer

$$E\left(\epsilon_i\right) = 0,$$

Then

$$\underset{\substack{k\times 1 \\ \text{bias}}}{\delta} = E\left(\widehat{\beta} - \beta\right) = E\left(\widehat{\beta}\right) - \beta = 0.$$

Proof: note

$$\begin{aligned} & E\left(\widehat{\beta} - \beta\right) \\ = \;& E\left[\left(X'X\right)^{-1}\left(X'\epsilon\right)\right] \\ = \;& \left(X'X\right)^{-1} X'E\left(\epsilon\right) = \left(X'X\right)^{-1} X'0 = 0. \end{aligned}$$

- look at (4): $E(\epsilon_i) = 0$ means that on average, the model does not over or understate y_i;
- $\delta = 0$ means that on average, the OLS does not always make the same mistake, it picks the true value of β;
- note that I do not need $N \to \infty$.

Variance Assume that

$$\underset{N\times N}{E(\epsilon\epsilon')} = \sigma^2 I_N \tag{5}$$

then the "variance of $\widehat{\beta} - \beta$", $Var\left(\widehat{\beta} - \beta\right)$

$$\underset{k\times k}{E\left[\left(\widehat{\beta} - \beta\right)\left(\widehat{\beta} - \beta\right)'\right]} = \sigma^2 \left(X'X\right)^{-1}.$$

Proof:

$$\begin{aligned}
& E\left[\left(\widehat{\beta} - \beta\right)\left(\widehat{\beta} - \beta\right)'\right] \\
= \; & E\left[\left(X'X\right)^{-1} X'\epsilon\epsilon' X \left(X'X\right)^{-1}\right] \\
= \; & \left(X'X\right)^{-1} X' E(\epsilon\epsilon') X \left(X'X\right)^{-1} \\
= \; & \left(X'X\right)^{-1} X' \sigma^2 I_N X \left(X'X\right)^{-1} \\
= \; & \sigma^2 \left(X'X\right)^{-1} \left(X'X\right) \left(X'X\right)^{-1} = \sigma^2 \left(X'X\right)^{-1}.
\end{aligned}$$

- the variance measures how big $\widehat{\beta} - \beta$ is: the smaller the better;
- the assumption $E(\epsilon\epsilon') = \sigma^2 I_N$ helps simplify the algebra. It is called *sphericity* and it means that
  - $E\left(\epsilon_i^2\right) = \sigma^2$, constant across units - *homoskedasticity*
  - $E(\epsilon_i \epsilon_j) = Cov(\epsilon_i, \epsilon_j) = 0$ - no cross correlation
- if we want to calculate $\sigma^2 \left(X'X\right)^{-1}$, then you need an estimate of σ^2. FYI this estimator exists and it is given by

$$\widehat{\sigma}^2 = \frac{\widehat{\epsilon}'\widehat{\epsilon}}{N - k}.$$

Gauss-Markov Theorem Under (4) and (5), OLS is BLUE (= Best Linear Unbiased Estimator).

Proof: let us construct another linear unbiased estimator

$$\widetilde{\beta} = \underset{k\times N}{C} y.$$

It must be unbiased, so

$$E\left(\widetilde{\beta}\right) = \beta$$

which requires

$$E\left(\widetilde{\beta}\right) = E\left(Cy\right) = \beta$$

so

$$\begin{aligned} & E\left(CX\beta + C\epsilon\right) \\ = \; & E\left(CX\beta\right) + E\left(C\epsilon\right) \\ = \; & CX\beta + CE\left(\epsilon\right) \\ = \; & CX\beta. \end{aligned}$$

In order to have no bias, I need

$$CX = I_k.$$

Now let's see

$$\begin{aligned} & \widetilde{\beta} - \beta \\ = \; & Cy - \beta \\ = \; & CX\beta + C\epsilon - \beta \\ = \; & \beta + C\epsilon - \beta \\ = \; & C\epsilon. \end{aligned}$$

I want to evaluate

$$\begin{aligned} & E\left[\left(\widetilde{\beta} - \beta\right)\left(\widetilde{\beta} - \beta\right)'\right] \\ = \; & E\left[C\epsilon\epsilon' C'\right] \\ = \; & CE\left[\epsilon\epsilon'\right] C' \\ = \; & \sigma^2 CC'. \end{aligned}$$

Define now the new matrices

$$\begin{aligned} D &= C - \left(X'X\right)^{-1} X' \\ C &= D + \left(X'X\right)^{-1} X' \end{aligned}$$

Then

$$\begin{aligned} & E\left[\left(\widetilde{\beta} - \beta\right)\left(\widetilde{\beta} - \beta\right)'\right] \\ = \; & \sigma^2 CC' \\ = \; & \sigma^2 \left[D + \left(X'X\right)^{-1} X'\right]\left[D + \left(X'X\right)^{-1} X'\right]' \\ = \; & \sigma^2 DD' + \sigma^2 \left(X'X\right)^{-1} X'X \left(X'X\right)^{-1} \\ & + \sigma^2 \left[DX\left(X'X\right)^{-1} + \left(X'X\right)^{-1} X'D'\right] \\ = \; & \sigma^2 DD' + \sigma^2 \left(X'X\right)^{-1} + \sigma^2 \left[DX\left(X'X\right)^{-1} + \left(X'X\right)^{-1} X'D'\right] \end{aligned}$$

But now note that

$$\begin{aligned} DX &= \left[C - (X'X)^{-1} X'\right] X \\ &= CX - (X'X)^{-1} X'X \\ &= I_k - I_k = 0, \end{aligned}$$

so the expression above simplifies

$$\begin{aligned} & E\left[\left(\widetilde{\beta} - \beta\right)\left(\widetilde{\beta} - \beta\right)'\right] \\ = \;& \sigma^2 DD' + \sigma^2 (X'X)^{-1} + \sigma^2 \left[DX (X'X)^{-1} + (X'X)^{-1} X'D'\right] \\ = \;& \sigma^2 DD' + \sigma^2 (X'X)^{-1} \\ = \;& \sigma^2 DD' + E\left[\left(\widehat{\beta} - \beta\right)\left(\widehat{\beta} - \beta\right)'\right] \end{aligned}$$

but $\sigma^2 DD'$ cannot be negative definite, so

$$E\left[\left(\widetilde{\beta} - \beta\right)\left(\widetilde{\beta} - \beta\right)'\right] \geq E\left[\left(\widehat{\beta} - \beta\right)\left(\widehat{\beta} - \beta\right)'\right]$$

which states that $\widetilde{\beta}$ is never better than OLS.

- recall OLS is unbiased;
- note that
  $$\widehat{\beta} - \beta = (X'X)^{-1} (X'\epsilon)$$
  i.e. the estimation error is a linear combination of the errors ϵ_i. Alternatively
  $$\widehat{\beta} = (X'X)^{-1} (X'y)$$
  is a linear combination of the regressors y_i. It is a computationally simple estimator.
- BEST (efficient) means that $E\left[\left(\widehat{\beta} - \beta\right)\left(\widehat{\beta} - \beta\right)'\right]$ is the smallest you can get, among linear and unbiased estimators: *if you want a better estimator, either you go nonlinear or you go biased*

Normality Assume

$$\epsilon \sim N\left(0, \sigma^2 I_N\right) \tag{6}$$

then

$$\begin{aligned} \widehat{\beta} - \beta &\sim N\left(0, \sigma^2 (X'X)^{-1}\right) \\ \widehat{\beta} &\sim N\left(\beta, \sigma^2 (X'X)^{-1}\right) \end{aligned}$$

- pretend $k = 1$, then I can do 2 things

– confidence intervals

$$\begin{aligned} \beta &\in \left(\widehat{\beta} \pm 1.96\sigma^2 \left(X'X\right)^{-1}\right) \\ & \left(\widehat{\beta} \pm 1.96\widehat{\sigma}^2 \left(X'X\right)^{-1}\right) \end{aligned}$$

where 1.96 is the 97.5 percentile of the normal. Then

$$P\left[\beta \in \left(\widehat{\beta} \pm 1.96\sigma^2 \left(X'X\right)^{-1}\right)\right] = 0.95,$$

i.e. I am quite confident that the true β will belong in that interval;

– the t-test. You can test for

$$H_0 : \beta = 0$$

which means that the corresponding regressors is useless (β is *insignificant*) because under H_0

$$\widehat{\beta} \sim N\left(0, \sigma^2 \left(X'X\right)^{-1}\right)$$

and then if you define

$$\frac{\widehat{\beta}}{\widehat{\sigma}^2 \left(X'X\right)^{-1}} = t_\beta$$

and decide that

$$|t_\beta| \begin{array}{ll} > 1.96 & H_0 \text{ false} \\ < 1.96 & H_0 \text{ true} \end{array}$$

- note I do not need $N \to \infty$. BUT, if ϵ is not normal, and I have $N \to \infty$, the property still holds.

Forecasting with OLS

Recall

$$\underset{N\times 1}{y} = \underset{\text{model}}{X \underset{k\times 1}{\beta}} + \epsilon,$$

recall

$$\begin{aligned} \widehat{\beta}^{OLS} &= \left(X'X\right)^{-1}\left(X'y\right) \\ \widehat{\beta} - \beta &= \left(X'X\right)^{-1}\left(X'\epsilon\right) \end{aligned}$$

and recall that

$$\begin{aligned} E\left(\widehat{\beta} - \beta\right) &= E\left(\widehat{\beta}\right) - \beta \equiv \underset{k\times 1}{\delta} = 0 \\ \underset{k\times k}{\underset{Var\left(\widehat{\beta}-\beta\right)}{E\left[\left(\widehat{\beta} - \beta\right)\left(\widehat{\beta} - \beta\right)'\right]}} &= \sigma^2 \left(X'X\right)^{-1} \end{aligned}$$

The first property, unbiasedness, really requires the more complicated condition (compared to $E(\epsilon) = 0$) called *weak exogeneity*

$$Cov(X, \epsilon) = 0,$$

which in general cannot be guaranteed if you omit regressors/variables. This is called *omitted variable bias*. If you want to see the algebra

$$y = X\beta + z\gamma + u$$

but you estimate, wrongly

$$y = X\beta + \underset{z\gamma+u}{\epsilon}$$

you then get

$$\begin{aligned}\widehat{\beta} - \beta &= (X'X)^{-1}(X'\epsilon) \\ &= (X'X)^{-1}X'(z\gamma + u) \\ &= (X'X)^{-1}X'z\gamma + (X'X)^{-1}X'u\end{aligned}$$

and it holds that

$$E\left[(X'X)^{-1}X'u\right] = 0,$$

but I do not know in general if

$$E\left[(X'X)^{-1}X'z\gamma\right] = 0,$$

because $Cov(X, z) = 0$ may not be true.

Let's talk about the variance of OLS

$$E\left[\left(\widehat{\beta} - \beta\right)\left(\widehat{\beta} - \beta\right)'\right] = \sigma^2 (X'X)^{-1},$$

which is the smallest attainable for an unbiased, linear estimator (Gauss-Markov theorem).

Recall that that the OLS estimator requires another important condition:

$$(X'X)^{-1}$$

must exists, i.e. $X'X$ must be invertible, or

$$rank\left(\underset{k\times k}{X'X}\right) = k.$$

This fails under some circumstances, including

1. certainly true when $k > N$ as is the case in

    (a) modern macroeconometrics, large VARs;

(b) asset pricing, where there are at least k =400 potential explanatory variables;

2. even when $k < N$, you may have *collinearity* (when regressors are very correlated with each other) - $rank\,(X'X) < k$

If OLS exists, however, let us use it to predict y via

$$\begin{aligned} \widehat{y} &= X\widehat{\beta}^{OLS} \\ &= \Big[\underbrace{X\,(X'X)^{-1}\,X'}_{P}\Big]y \\ &= Py. \end{aligned}$$

Let us evaluate the quality of these predictions

$$\underset{N\times 1}{y-\widehat{y}}$$

by computing

$$\begin{aligned} E\,(y-\widehat{y}) &= E\,(y) - E\,(\widehat{y}) \\ &= E\,(X\beta+\epsilon) - E\left(\left[X\,(X'X)^{-1}\,X'\right]y\right) \\ &= X\beta + E\,(\epsilon) - \left[X\,(X'X)^{-1}\,X'\right]E\,(y) \\ &= X\beta + \underbrace{E\,(\epsilon)}_{=0} - \left[X\,(X'X)^{-1}\,X'\right][X\beta+E\,(\epsilon)] \\ &= X\beta - X\,(X'X)^{-1}\,X'X\beta = X\beta - X\beta = 0 \end{aligned}$$

i.e. predictions are *unbiased.*

Also

$$Var\,(y-\widehat{y}) = E\,\big[\underbrace{(y-\widehat{y})\,(y-\widehat{y})'}_{N\times N}\big] = \sigma^2\Big[\underbrace{I_N - \underbrace{X\,(X'X)^{-1}\,X'}_{P}}_{M}\Big].$$

A typical summary of these two properties is the Mean Squared Error

$$\begin{aligned} & MSE\left(\underset{y-\widehat{y}}{\widehat{y}}\right) \\ = \;& bias^2 + variance \\ = \;& \underbrace{[\underbrace{E\,(y-\widehat{y})}_{N\times 1}][\underbrace{E\,(y-\widehat{y})}_{1\times N}]'}_{N\times N} + E\,\big[(y-\widehat{y})\,(y-\widehat{y})'\big] \end{aligned}$$

which is never negative. We would like it to be small, but how much is it? We could show that

$$\begin{aligned} & MSE(\widehat{y}) \\ = \; & \sigma^2 I_N + \sigma^2 (X'X)^{-1} \\ = \; & \underbrace{\sigma^2}_{Var(\epsilon_i)} I_N + X \left[\underbrace{MSE\left(\widehat{\beta}^{OLS}\right)}_{\sigma^2(X'X)^{-1}} \right] X' \end{aligned}$$

so, this is big when

1. σ^2 is big: it means that ϵ is big, i.e. bad model;
2. a bad estimator is used: $MSE\left(\widehat{\beta}^{OLS}\right)$ big means bad predictions.

Does OLS have shortcomings?

1. it may not exist, when $k > N$
2. look at the predictive point of view: $MSE\left(\widehat{\beta}^{OLS}\right)$ boils down to

$$MSE\left(\widehat{\beta}^{OLS}\right) = \underbrace{\delta\delta'}_{=0} + \sigma^2 (X'X)^{-1}$$

   so it only depends on $\sigma^2 (X'X)^{-1}$. But this can be huge, whenever $(X'X)^{-1}$ is huge/inflated, which happens when

   (a) k is huge, in general;

   (b) when regressors are correlated.

3. this also comes with interpretability problems. OLS can give nonsense results when $(X'X)^{-1}$ is huge/inflated. No algebra, but when $(X'X)^{-1}$ is huge/inflated, you end up having

   (a) $\widehat{\beta}_j$ will look huge and with maybe the wrong sign;

   (b) has a massive standard error (=very large confidence intervals);

   (c) turns out to be actually insignificant.

What can we say from this and about this?
We have seen that

1. OLS has low bias (=0, it cannot be better!), as long as you do not omit regressors

   (a) I can expect a (big) bias if k is small;

2. OLS has low variance (=Gauss-Markov theorem), but only when $(X'X)^{-1}$ is NOT huge/inflated. But this entails that

   (a) I can expect a big variance if k is large.

This is the *bias/variance tradeoff:* OLS suffers from it, like all other estimators.

Improving the OLS: the Ridge

To improve the OLS, we may want to play with the bias/variance tradeoff. We may be able to reduce MSE of predictions by e.g. reducing the variance of OLS. But this is NOT possible: OLS has the smallest variance among unbiased estimators.

So, how about ... we accept some bias, and hope to reduce the variance so as to have an overall reduction of MSE?

Recall

$$\underset{N\times 1}{y} = X \underset{k\times 1}{\beta} + \epsilon.$$

One idea could be as follows: there are regressors which are significant ($\beta_j \neq 0$), but are "useless" (β_j is small). How about I throw them away?

- I omit a relevant variable, so probably I'll have a bias;
- but now k is smaller, so maybe $(X'X)^{-1}$ is deflated, i.e. lower variance.

Given that

- "useless" regressors have small but nonzero β_j
- you drop them, meaning that you force $\beta_j = 0$

we call this procedure *shrinkage*.

How does this work mathematically? The OLS minimised the RSS, i.e.

$$\begin{aligned} RSS &= \sum_{i=1}^{N} \widehat{\epsilon}_i^2 \\ &= \widehat{\epsilon}'\widehat{\epsilon} \\ &= \left(y - X\widehat{\beta}\right)'\left(y - X\widehat{\beta}\right) \\ &= \left\|y - X\widehat{\beta}\right\|_2^2 \end{aligned}$$

Our new estimator, which is based on shrinkage, minimises a different loss function. It minimises

$$\begin{aligned}
\widehat{\beta}^{Ridge} &= \arg\min_{\beta}\left(\underbrace{\|y - X\beta\|_2^2}_{\text{completeness}} + \lambda \underbrace{\|\beta\|_2^2}_{\text{complexity}}\right), \\
\|\beta\|_2^2 &= \sum_{j=1}^{k}\beta_j^2
\end{aligned}$$

where $\lambda \geq 0$ is a user-defined tuning parameter.

This expression deserves some comments

1. the loss function is the same as OLS, containing $\|y - X\beta\|_2^2$, but there is a further component, $\lambda\|\beta\|_2^2$
   (a) adding an extra regressors means having an extra component β_j in β. This comes at a cost: you have to pay λ (which is the unit cost per regressor);
   (b) the OLS never demanded this;
   (c) this should create an incentive to use only the regressors we do need: more regressors will reduce $\|y - X\beta\|_2^2$, but this benefit will have a cost, given by $\lambda\|\beta\|_2^2$;
2. the key parameter seems to be $\lambda \in [0, \infty]$
   (a) when $\lambda = 0$, my penalty is only $\|y - X\beta\|_2^2$, and therefore $\widehat{\beta}^{Ridge} = \widehat{\beta}^{OLS}$;
   (b) when $\lambda = \infty$, each regressor will have an infinite, unaffordable price: you will end up with a model with a constant only: you give up on having a model
   (c) in between, we are balancing two ideas. Having all the useful regressors, and having no useless ones.
3. practicalities:
   (a) the penalty makes sense but it is unfair, as it is sensitive to the unit of measurement of the regressors

   $$y_i = \beta_1 \underbrace{x_{1,i}}_{\$} + \beta_2 \underbrace{x_{2,i}}_{1000\$} + error$$

   so, when you know/suspect that variables are not on the same scale, we typically standardise regressors prior to using Ridge, i.e.

   $$\begin{aligned}
   \widetilde{x}_{1,i} &\rightarrow \frac{x_{1,i}}{\sqrt{Var(x_{1,i})}} \\
   \widetilde{x}_{2,i} &\rightarrow \frac{x_{2,i}}{\sqrt{Var(x_{2,i})}}
   \end{aligned}$$

   and now $\widetilde{x}_{1,i}$ and $\widetilde{x}_{2,i}$ are on the same scale.

Let us derive the Ridge. Similar passages as OLS and remember

$$\begin{aligned}\frac{\partial}{\partial x} Ax &= A' \\ \frac{\partial}{\partial x} x'Ax &= (A + A')\,x \underset{\text{if A symmetric}}{=} 2Ax\end{aligned}$$

Now we have

$$\begin{aligned}& \min_{\beta} \left(\|y - X\beta\|_2^2 + \lambda \|\beta\|_2^2 \right) \\ = & \min_{\beta} \left[(y - X\beta)' (y - X\beta) + \lambda \beta' \beta \right]\end{aligned}$$

I need to evaluate

$$\begin{aligned}& \frac{\partial}{\partial \beta} \left[(y - X\beta)' (y - X\beta) + \lambda \beta' \beta \right] \\ = & \frac{\partial}{\partial \beta} \left[y'y + \beta' X' X \beta - 2y' X \beta + \lambda \beta' \beta \right] \\ = & 0 + 2X'X\beta - 2X'y + 2\lambda I_k \beta = 0\end{aligned}$$

and I obtain

$$[X'X + \lambda I_k]\,\beta = X'y$$

whence

$$\widehat{\beta}^{Ridge} = [X'X + \lambda I_k]^{-1} X'y.$$

We can study Ridge and its properties by using

$$\widehat{\beta}^{Ridge} = [X'X + \lambda I_k]^{-1} X' (X\beta + \epsilon).$$

Theorem 1 *Under the assumptions of the OLS*

$$\begin{aligned}E\left(\widehat{\beta}^{Ridge}\right) &= [X'X + \lambda I_k]^{-1} X'X\beta \neq \beta \\ Var\left(\widehat{\beta}^{Ridge} - \beta\right) &= \sigma^2 [X'X + \lambda I_k]^{-1} [X'X] [X'X + \lambda I_k]^{-1}\end{aligned}$$

So, Ridge is biased! With bias

$$\delta = \left([X'X + \lambda I_k]^{-1} X'X - I_k \right) \beta.$$

Some facts

1. $\delta = 0$ only when $\lambda = 0$ (of course, in that case we have the OLS!);

    (a) we have

    $$\frac{\partial}{\partial \lambda} (\delta \delta') > 0$$

    i.e., if λ increases, there is more bias (because you omit more regressors)

2. the variance is more difficult to study but
   (a) it is smaller than OLS
   $$\sigma^2 \left[X'X + \lambda I_k\right]^{-1} \left[X'X\right] \left[X'X + \lambda I_k\right]^{-1} < \sigma^2 \left[X'X\right]^{-1}$$
   which I should hope, since otherwise Ridge is worse than OLS and pointless
   (b) also
   $$\frac{\partial}{\partial \lambda} Var\left(\widehat{\beta}^{Ridge} - \beta\right) < 0.$$
3. this seems to suggest that there is an optimal λ, say $\widetilde{\lambda}$, such that
   $$\widetilde{\lambda} = \arg\min_{\lambda} MSE\left(\widehat{\beta}^{Ridge}\right).$$
4. Some further practical notes
   (a) if all β_js are quite big, what happens? Does the Ridge still help? Yes! Shrinking is still helpful, although less than if some β_j is very small;
   (b) suppose in your model some $\beta_j = 0$: not small, really 0. This means that the corresponding regressor is useless AND this has a nice economic interpretation:
      i. when you use OLS, and you find $\beta_j = 0$, this has an interpretation
         A. and you drop the regressor and run your model again without it: this called **variable selection**;
      ii. the Ridge will never tell you that $\beta_j = 0$: every β_j is estimated as a nonzero number, just very small
         A. predictions will be great, and better than OLS
         B. but you cannot do variable selection: you'll get rid of regressors. Interpretability of a Ridge regression is lost, even if predictions are great.

The LASSO

A brief revision, as usual

$$\begin{aligned} \underset{N\times 1}{y} &= X \underset{k\times 1}{\beta} + \epsilon \\ y_i &= \beta_1 + \sum_{j=2}^{k} \beta_j x_{j,i} + \epsilon_i = x_i'\beta + \epsilon_i \end{aligned}$$

We want to

- estimate β, obtaining $\widehat{\beta}$
- for at least two purposes
  - interpretation;
  - prediction;
- let's say we care mainly about predictions. There will be an error

$$\begin{aligned} \widehat{y}_i &= x_i'\widehat{\beta} \\ y_i - \widehat{y}_i &= \epsilon_i + x_i'\left(\beta - \widehat{\beta}\right) \end{aligned}$$

  whose magnitude we measure as

$$MSE\left(y_i - \widehat{y}_i\right) = Var\left(\epsilon_i\right) + x_i' MSE\left(\widehat{\beta}\right) x_i$$

  where we know that

$$\begin{aligned} MSE\left(\widehat{\beta}\right) &= bias^2 + variance \\ &= \delta\delta' + E\left[\left(\widehat{\beta} - E\left(\widehat{\beta}\right)\right)\left(\widehat{\beta} - E\left(\widehat{\beta}\right)\right)'\right] \end{aligned}$$

  where

$$\delta = E\left(\widehat{\beta}\right) - \beta.$$

- we know that
  - OLS is great because it can be unbiased, but
    * this requires k large enough - i.e. many regressors to avoid omitted variable bias;
    * this will inflate the variance though, and OLS may not exist $(k > N)$ - think about $(X'X)^{-1}$;
  - OLS is great because if $\beta_j = 0$, I can find out through a very standard t-test
    * I can knock off $x_{j,i}$ (variable selection);
    * I can interpret this.
  - Ridge is great because we can tune bias and variance to our advantage
    * it shrinks all the $\widehat{\beta}_j$ towards zero compared to OLS

$$\left|\widehat{\beta}_j^{ridge}\right| \leq \left|\widehat{\beta}_j^{OLS}\right|$$

    * irrelevant/less than relevant regressors are massively downplayed;
    * it delivers better predictions than OLS, but

* it will never force a $\widehat{\beta}_j = 0$ when $\beta_j = 0$: there is no way of testing for $\beta_j = 0$, $\widehat{\beta}_j \neq 0$ and therefore interpretability is lost.

To get the best of both worlds, statisticians created the LASSO = Least Absolute Selection and Shrinkage Operator. Recall the Ridge

$$\min_{\beta}\left(\|y - X\beta\|_2^2 + \lambda\|\beta\|_2^2\right),$$

where

$$\|\beta\|_2^2 = \sum_{j=1}^{k}\beta_j^2$$

the L_2-norm makes the problem very tractable.

The LASSO works in a similar way:

$$\min_{\beta}\left(\underbrace{\|y - X\beta\|_2^2}_{RSS} + \underbrace{\lambda}_{\in[0,\infty]}\underbrace{\|\beta\|_1}_{penalty}\right),$$

where

$$\|\beta\|_1 = \sum_{j=1}^{k}|\beta_j|.$$

Let us solve the minimisation problem. We have

$$\begin{aligned}
\widehat{\beta}^{lasso} &= \arg\min_{\beta}\left(\|y - X\beta\|_2^2 + N\frac{\lambda}{N}\|\beta\|_1\right)\\
&= \arg\min_{\beta}\left(\|y - X\beta\|_2^2 + N\widetilde{\lambda}\|\beta\|_1\right)
\end{aligned}$$

Let us consider a simple case where

$$X'X = NI_k$$

and study

$$\begin{aligned}
&\arg\min_{\beta}\frac{1}{N}\left(\|y - X\beta\|_2^2 + N\widetilde{\lambda}\|\beta\|_1\right)\\
=\ &\arg\min_{\beta}\left(\frac{1}{N}\|y - X\beta\|_2^2 + \widetilde{\lambda}\|\beta\|_1\right)\\
=\ &\arg\min_{\beta}\left(\frac{1}{N}\sum_{i=1}^{N}\left(y_i - x_i'\beta \pm x_i'\widehat{\beta}^{OLS}\right)^2 + \widetilde{\lambda}\sum_{j=1}^{k}|\beta_j|\right)\\
=\ &\arg\min_{\beta}\left(\frac{1}{N}\sum_{i=1}^{N}\left(\left(y_i - x_i'\widehat{\beta}^{OLS}\right) - \left(x_i'\beta - x_i'\widehat{\beta}^{OLS}\right)\right)^2 + \widetilde{\lambda}\sum_{j=1}^{k}|\beta_j|\right)\\
=\ &\arg\min_{\beta}\left(\frac{1}{N}\sum_{i=1}^{N}\left(y_i - x_i'\widehat{\beta}^{OLS}\right)^2 + \frac{1}{N}\sum_{i=1}^{N}\left(x_i'\beta - x_i'\widehat{\beta}^{OLS}\right)^2 + \widetilde{\lambda}\sum_{j=1}^{k}|\beta_j|\right)
\end{aligned}$$

and there are no cross products because you can check that

$$\sum_{i=1}^{N} \left(y_i - x_i'\widehat{\beta}^{OLS}\right)\left(x_i'\beta - x_i'\widehat{\beta}^{OLS}\right) = 0$$

Now

$$\begin{aligned}
& \arg\min_{\beta} \left(\frac{1}{N} \sum_{i=1}^{N} \left(y_i - x_i'\widehat{\beta}^{OLS}\right)^2 + \frac{1}{N} \sum_{i=1}^{N} \left(x_i'\beta - x_i'\widehat{\beta}^{OLS}\right)^2 + \widetilde{\lambda} \sum_{j=1}^{k} |\beta_j| \right) \\
= \; & \arg\min_{\beta} \left(\frac{1}{N} \sum_{i=1}^{N} \left(x_i'\beta - x_i'\widehat{\beta}^{OLS}\right)^2 + \widetilde{\lambda} \sum_{j=1}^{k} |\beta_j| \right) \\
= \; & \arg\min_{\beta} \left(\frac{1}{N} \left\| X\left(\beta - \widehat{\beta}^{OLS}\right) \right\|_2^2 + \widetilde{\lambda} \sum_{j=1}^{k} |\beta_j| \right) \\
= \; & \arg\min_{\beta} \left(\frac{1}{N} \left(\widehat{\beta}^{OLS} - \beta\right)' \underbrace{X'X}_{=NI_k} \left(\widehat{\beta}^{OLS} - \beta\right) + \widetilde{\lambda} \sum_{j=1}^{k} |\beta_j| \right) \\
= \; & \arg\min_{\beta} \left(\left(\widehat{\beta}^{OLS} - \beta\right)' \left(\widehat{\beta}^{OLS} - \beta\right) + \widetilde{\lambda} \sum_{j=1}^{k} |\beta_j| \right)
\end{aligned}$$

$$\begin{aligned}
= \; & \arg\min_{\beta} \left(\left(\widehat{\beta}^{OLS} - \beta\right)' \left(\widehat{\beta}^{OLS} - \beta\right) + \widetilde{\lambda} \sum_{j=1}^{k} |\beta_j| \right) \\
= \; & \arg\min_{\beta} \left(\sum_{j=1}^{k} \left|\widehat{\beta}_j^{OLS} - \beta_j\right|^2 + \widetilde{\lambda} \sum_{j=1}^{k} |\beta_j| \right)
\end{aligned}$$

This problem can be studied one j at a time, i.e.

$$\arg\min_{\beta_j} \left(\left|\widehat{\beta}_j^{OLS} - \beta_j\right|^2 + \widetilde{\lambda} |\beta_j| \right)$$

So, I need to separate two cases

- $\beta_j > 0$, we have

$$\arg\min_{\beta_j} \left(\left|\widehat{\beta}_j^{OLS} - \beta_j\right|^2 + \widetilde{\lambda}\beta_j \right)$$

  I'll take the derivative and set to zero

$$\begin{aligned}
-2\left(\widehat{\beta}_j^{OLS} - \beta_j\right) + \widetilde{\lambda} &= 0 \\
\widehat{\beta}_j^{lasso} &= \widehat{\beta}_j^{OLS} - \widetilde{\lambda}/2
\end{aligned}$$

but this is true only for $\widehat{\beta}_j^{OLS} \geq \widetilde{\lambda}/2$; otherwise, you will have to truncate, i.e.

$$\widehat{\beta}_j^{lasso} = 0$$

- $\beta_j < 0$, we have

$$\arg\min_{\beta_j} \left(\left| \widehat{\beta}_j^{OLS} - \beta_j \right|^2 - \widetilde{\lambda}\beta_j \right)$$

  I'll take the derivative and set to zero

$$\begin{aligned} -2\left(\widehat{\beta}_j^{OLS} - \beta_j\right) - \widetilde{\lambda} &= 0 \\ \widehat{\beta}_j^{lasso} &= \widehat{\beta}_j^{OLS} + \widetilde{\lambda}/2 \end{aligned}$$

  but this is true only for $\widehat{\beta}_j^{OLS} \leq -\widetilde{\lambda}/2$; otherwise, you will have to truncate, i.e.

$$\widehat{\beta}_j^{lasso} = 0$$

Hence I have

$$\widehat{\beta}_j^{lasso} = \begin{cases} \widehat{\beta}_j^{OLS} - \widetilde{\lambda}/2 & \widehat{\beta}_j^{OLS} > \widetilde{\lambda}/2 \\ 0 & -\widetilde{\lambda}/2 < \widehat{\beta}_j^{OLS} < \widetilde{\lambda}/2 \\ \widehat{\beta}_j^{OLS} + \widetilde{\lambda}/2 & \widehat{\beta}_j^{OLS} < -\widetilde{\lambda}/2 \end{cases}$$